

State Space Tree

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State Space Tree

- A State Space Tree is a tree representation used to explore all possible states or solutions in problems related to backtracking, branch and bound, and search algorithms. It is commonly used in problems like constraint satisfaction, and pathfinding.



State Space Tree

- 1- initial state
- 2- Goal State
- 3- operators
- 4- movements
- 5- stopping condition
- 6- State Variable

Example X-O Game :

1- initial state

empty 3x3 board

2- Goal state

To have the same symbol at (row or col or diagonal)

3-operators

X – O

Example X-O Game :

4-Movements

Mark empty space

5-Stopping condition

Goal state / no empty spaces left

6-State Variable

2D list [3*3]



Sheet 1 :

You have a 4 x 4 chess board it is required to put 4 queens on the board such that no queen attacks another. Define the initial and goal states, state variables, operators, and draw the state space tree for the given problem

1- initial state

empty 4x4 board

2- Goal state

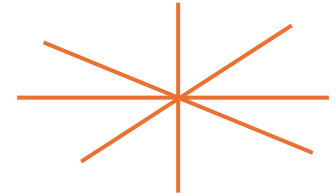
To put 4 queens on the board such that no queen attacks another

3-operators

4 Queens

4-Movements

Up , down , right , left , right diagonal , left diagonal



5-Stopping condition

Goal state / cannot put the 4th queen

6-State Variable

2D list [4*4]

Q	XX	XX	XX
XX	XX		
XX		XX	
XX			XX

XX	Q	XX	XX
XX	XX	XX	
	XX		XX
	XX		

XX	XX	Q	XX
	XX	XX	XX
XX		XX	
		XX	

XX			XX
	XX		XX
		XX	XX
XX	XX	XX	Q

Q	XX	XX	XX
XX	XX	Q	XX
XX	XX	XX	XX
XX		XX	XX

XX	Q	XX	XX
XX	XX	XX	Q
	XX	XX	XX
	XX		XX

XX	XX	Q	XX
Q	XX	XX	XX
XX	XX	XX	
XX		XX	

XX	Q	XX	XX
XX	XX	XX	XX
	XX	XX	XX
XX	XX	XX	Q

Q	XX	XX	XX
XX	XX	Q	XX
XX	XX	XX	XX
XX	Q	XX	XX

XX	Q	XX	XX
XX	XX	XX	Q
Q	XX	XX	XX
XX	XX		XX

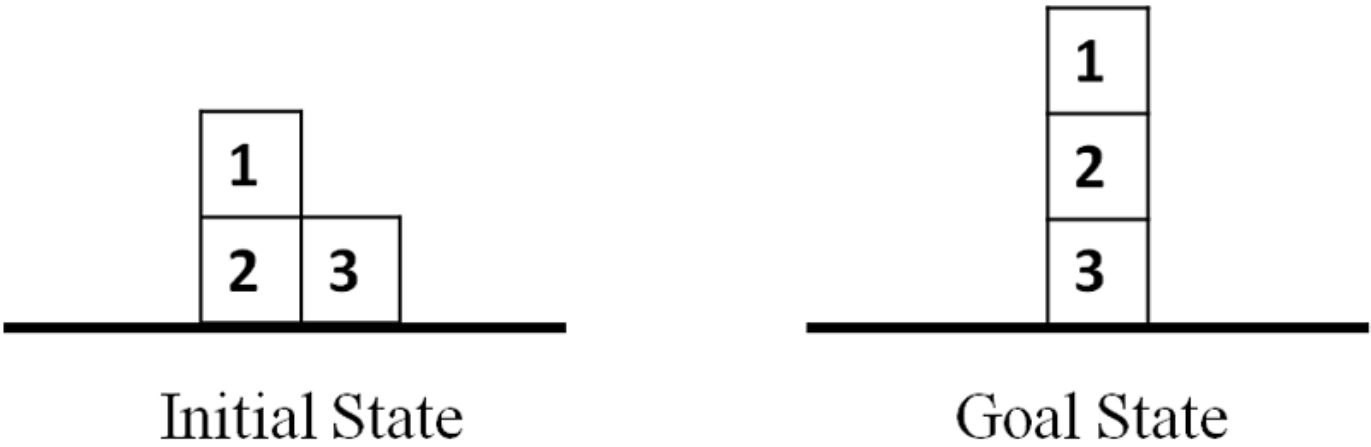
XX	XX	Q	XX
Q	XX	XX	XX
XX	XX	XX	Q
XX		XX	XX

XX	Q	XX	XX
XX	XX	XX	XX
Q	XX	XX	XX
XX	XX	XX	Q

XX	Q	XX	XX
XX	XX	XX	Q
Q	XX	XX	XX
XX	XX	Q	XX

XX	XX	Q	XX
Q	XX	XX	XX
XX	XX	XX	Q
XX	Q	XX	XX

The figure below shows a problem consisting of a table and 3 blocks. The legal operations are to move block X onto block Y if Y is a top block or to move X onto the table. Draw the state space tree for the given problem.



operations :

m(1,2)

m(1,3)

m(1,t)

m(2,1)

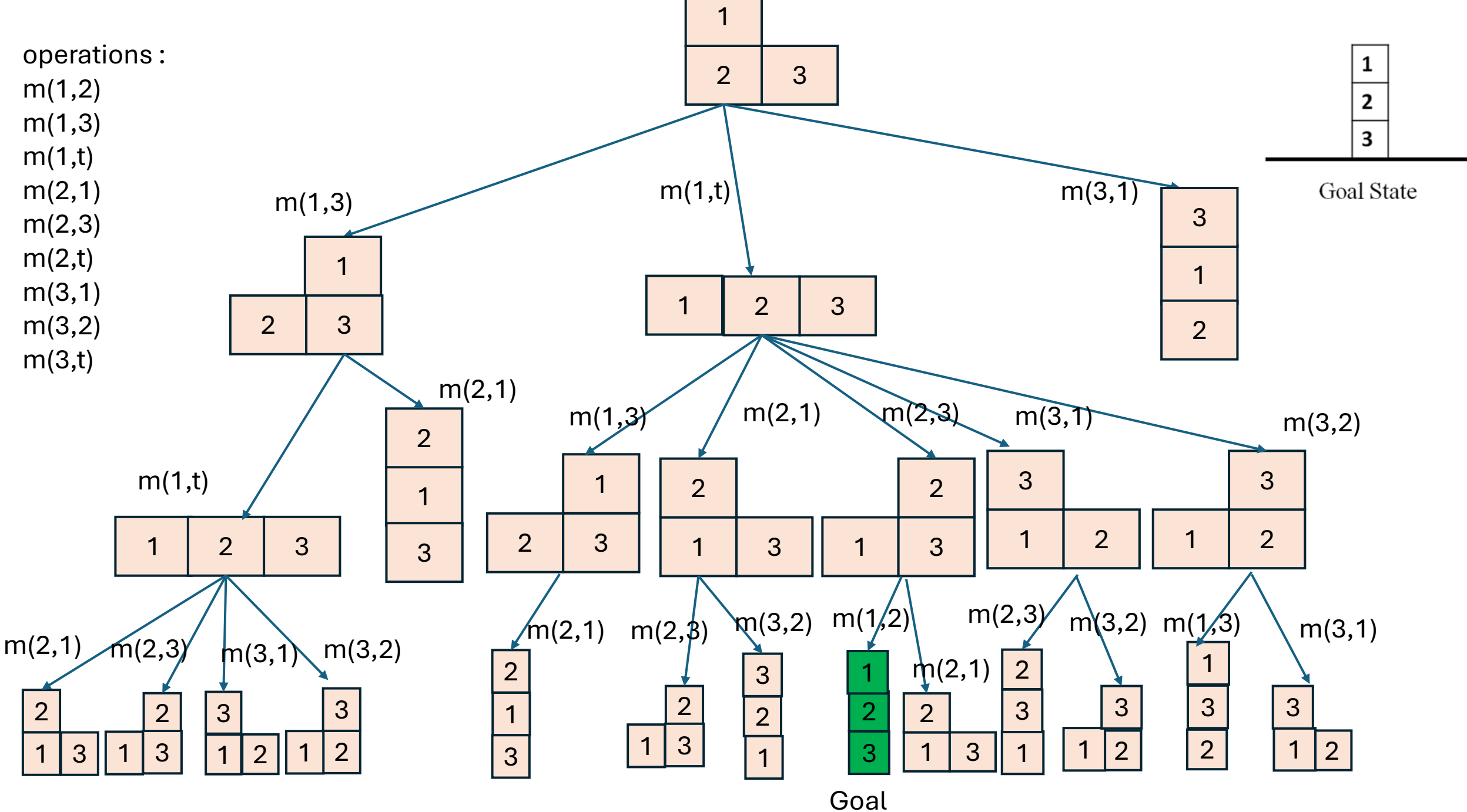
m(2,3)

m(2,t)

m(3,1)

m(3,2)

m(3,t)



- Given the initial and goal states, draw the state space tree for the following 8 puzzle problem.

2	1	4
3	8	7
5	6	

Initial State

1	2	3
4	5	6
7	8	

Goal State

operations :

mL
mU
mR
mD

2	1	4
3	8	7
5	6	

1	2	3
4	5	6
7	8	

Goal State

mL

2	1	4
3	8	7
5		6

mU

2	1	4
3	8	
5	6	7

mL

2	1	4
3	8	7
	5	6

mU

2	1	4
3		7
5	8	6

mL

2	1	4
3		8
5	6	7

mU

2	1	
3	8	4
5	6	7

Allowed moves in order are (left , up , right , down)

- You have 3 water jugs whose capacities are 10, 7 and 3 liters respectively. The first jug is full and the other 2 jugs are empty. It is required to put 5 liters in the first and 5 liters in the second. (Note: no additional source of water is available and no tool for measuring). Define the initial and goal states, state variables, operators, and draw the state space tree for the given problem.

operations :

A -> B

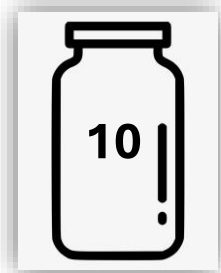
A -> C

B -> A

B -> C

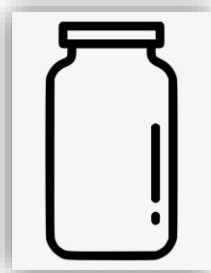
C -> A

C -> B



10L

A



7L

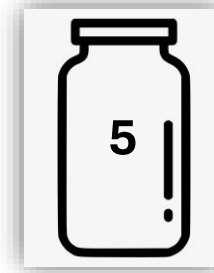
B



3L

C

Initial State



10L

A



7L

B



3L

C

Goal State